

Enhancement of viewer comfort in stereoscopic viewing: parallax adjustment

Janusz Konrad

INRS-Télécommunications, Institute National de la Recherche Scientifique
16 Place du Commerce, Verdun, QC, Canada, H3E 1H6
konrad@inrs-telecom.quebec.ca

ABSTRACT

One of the major deficiencies of stereoscopic visualization, viewer discomfort, can be caused by the non-robustness of human perception (hyper-sensitivity to 3-D) or by excessive 3-D cues in the viewed images. In order to minimize this discomfort, the amount of parallax (or "3D-ness") within each stereo pair needs to be reduced. Similarly to the case of "continuous look-around", parallax adjustment requires the knowledge of images from virtual cameras. In the case of parallel geometry, the virtual cameras are located on the line between the true cameras. Since in a general scenario no constraint should be posed on the complexity of the viewed scene, 3-D modeling techniques cannot be used. We evaluate the usefulness of parallax adjustment using two view reconstruction methods based on disparity-compensated linear interpolation: a quadtree method with block splitting adapted to object boundaries and a pixel based (dense) method. For all, but most complex, stereoscopic images tested (ITU-R 601 from CCETT and NHK) both algorithms performed very well, especially the pixel-based approach. In terms of the overall usefulness of parallax adjustment, the initial tests have shown a very favorable viewer response; the perceived depth was judged to vary smoothly from zero (one virtual camera) through natural 3-D (true cameras) to exaggerated 3-D (virtual cameras further apart than the true cameras - extrapolation). The adjustment was convincing although not completely free of distortions.

Keywords: Stereoscopic viewing, adjustment of parallax, intermediate view reconstruction, disparity estimation

1. INTRODUCTION

The current transition to digital storage and transmission of visual information opens new challenges but also new opportunities for stereoscopic and 3-D imaging systems. The transition to the digital medium is a clear challenge as new algorithms need to be developed for:

1. digital compression of stereoscopic and multi-view data to meet channel and/or storage capacity constraints,
2. signal processing to enhance user comfort (eye strain) and system functionality.

However, by addressing these challenges new opportunities will arise. For example, algorithms developed for one particular application are likely to find use in other applications. Moreover, new applications, never-imagined before will almost certainly emerge from such algorithms. One such example, in its early infancy today, is the generation of special 3-D effects from natural images rather than by means of computer graphics.

To date, a fair amount of research and development has been devoted to the compression of stereoscopic and multi-view images. Much less has been done, however, in terms of the viewer comfort. Yet, one of the major deficiencies of stereoscopic visualization remains to be viewer discomfort, not only due to the awkwardness of the viewing devices (e.g., eyewear), but also due to the eye strain. Such a strain is usually caused by the non-robustness of human perception and/or excessive 3-D cues in the presented images. By the non-robustness I mean the fact that among viewers there exists a very significant variation in sensitivity to stereoscopic cues.¹ Therefore, while some viewers have no problem with the fusion of a particular stereo image, others may be unable to do so or may feel eye strain. The problem may be further compounded by the presence of excessive 3-D cues in the viewed images. Although viewers may experience no fusion problems with properly-acquired stereoscopic images, i.e., with moderate parallax, significant difficulties may arise when parallax is excessive. For images acquired by a trained operator of a

stereoscopic camera, this should occur infrequently. However, if the images are acquired under conditions optimized for one particular viewing environment (e.g., TV screen), and then are presented in another environment (e.g., large-screen cinema), viewer discomfort may occur.

In order to minimize viewer discomfort, the amount of parallax within each stereo pair needs to be adjusted, usually by bringing the cameras closer together. However, in a scenario when a manipulation of true cameras is not possible, *virtual* cameras need to be manipulated by means of signal processing. One can also imagine a scenario where the amount of parallax is too small; to enhance the viewing experience it may need to be increased. This happens when images optimized for cinema are shown on TV screen. Since the disparities on a large screen (e.g., 10m) usually do not exceed 6-7cm to avoid eye divergence, they become very small when images are scaled down to a TV screen.

The virtual camera manipulation can be performed *via* disparity estimation, to establish correspondence between pixels in both images of a stereo pair, followed by disparity-compensated interpolation, to reconstruct the virtual views. The overall mechanism is identical to that used in *intermediate view reconstruction* (IVR) for continuous look-around and reported elsewhere in the literature.²⁻⁶ Another mechanism for view reconstruction is based on 3-D scene modeling,⁷ however thus far it has been demonstrated on images acquired in simple environments only; arbitrary scenes are too complex to be modeled today.

In this paper, I will discuss in more detail a parallax adjustment algorithm developed earlier in our laboratory and its recent extensions. Our results to date are very encouraging and indicate that good-quality 3-D adjustment is possible, perhaps with the exception of the most complex images. Therefore, it does not seem unlikely that a future 3-D TV or computer screen will have a "3D-ness" or "depthness" knob similar to the "brightness" and "contrast" adjustments used today.

2. APPROACH

Similarly to the case of continuous look-around (view adaptation),^{2,3} parallax adjustment requires the knowledge of images from virtual cameras. Clearly, both require intermediate view reconstruction.

To compute the intermediate views we use signal processing techniques rather than those based on 3-D modeling because:

1. 3-D modeling is imprecise for complex scenes,
2. in the intended applications camera baseline (distance between optical axes) is close to the interocular in humans (about 6-7cm) and therefore the range of disparities is rather limited,
3. the camera convergence angle in such applications is very small (at most 2-3 degrees), thus limiting geometric distortions (keystoning) as well as the disparity range.

In the case of parallel camera geometry, the "virtual" cameras, at which the reconstruction will take place, are located on the line passing through the optical centers of the true cameras and have optical axes parallel to the left and right camera axes (Fig. 1). If a pair of "virtual" cameras is located between the true cameras, a parallax reduction takes place and the depth in the perceived 3-D image is compressed (Fig. 1.a). On the other hand, if a pair of "virtual" cameras is outside of the true cameras, then a parallax increase results (Fig. 1.b); the perceived depth in the 3-D image is stretched.

3. ALGORITHMS

Below, the reconstruction of intermediate views is performed in two steps. First, a correspondence between the two images is established *via* disparity estimation. Then, using the estimated disparities, a disparity-compensated linear interpolation is performed. I will compare a block-based approach developed by us earlier with an extension of an established pixel-based method.

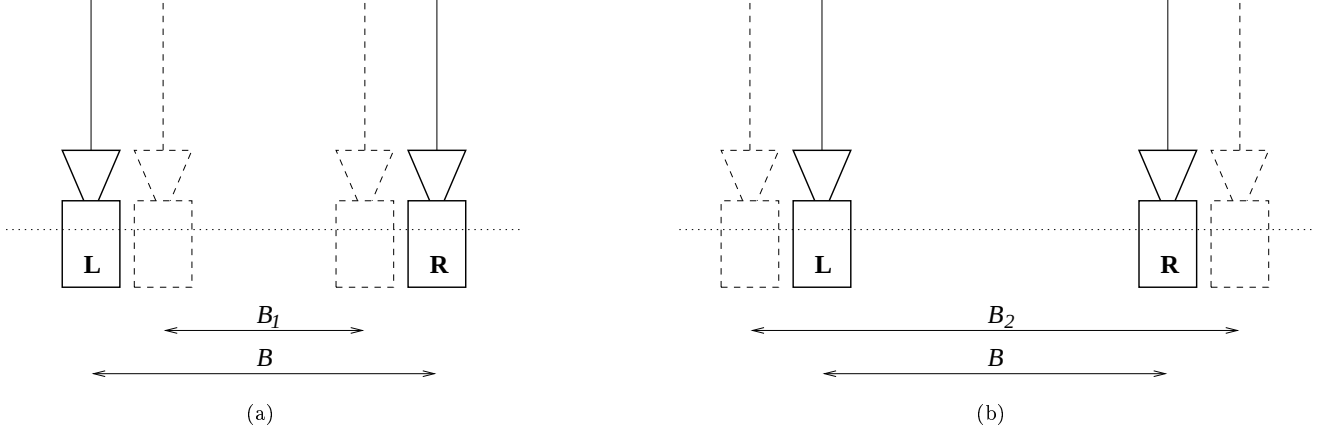


Figure 1. Location of virtual cameras (dashed lines) with respect to true cameras (solid lines) in the case of parallel camera geometry: (a) parallax reduction; (b) parallax increase.

3.1. Block-based IVR

In the block-based IVR method, it is assumed that a single disparity vector is assigned to all pixels within a rectangular block. Since disparity is inversely proportional to depth, this model translates to a constant depth within each block. Although this may be a reasonable approximation of depth within an object, this model fails at object boundaries due to depth discontinuities usually present there. Therefore, we have developed a robust method for quadtree splitting of disparity blocks with adaptation based on intensity mismatches (disparity-compensated prediction error) and robust statistics.⁶ The method is robust and quite insensitive to the parameters (thresholds) used. It assures splitting of blocks at most object boundaries and dramatically reduces reconstruction errors in those areas. In addition to the quadtree splitting, the method takes advantage of the following improvements:

1. spatial disparity smoothness constraint (regularization) between neighboring blocks to force local consistency of disparities within an object,
2. robust measure of intensity matching error (disparity compensation),^{8,9}
3. use of both color and luminance to compute the matching error, although the improvement due to the use of color is marginal,
4. use of luminance- and chrominance-balanced images¹⁰ to compensate for mismatched camera characteristics.

Details of this algorithm as well as examples of improvements induced by the above extensions can be found in our previous work.⁶

3.2. Pixel-based IVR

Although the block-based model with quadtree block division performs quite well, as will be discussed later, we have observed that the smallest block size for which reliable splitting decisions can be made is 8×8 pixels thus giving the finest divisions at the level of 4×4 pixels. This turns out to be insufficient precision for accurate rendering of object boundaries. Therefore, we have studied dense methods where a disparity vector is estimated at each pixel.

Let $I_L(x, y)$ and $I_R(x, y)$ denote intensities of the left and right images at spatial position (x, y) , and let $[u(x, y), v(x, y)]^T$ denote a disparity vector at the same position. We make the following, not uncommon, assumptions about the images and disparities:

1. the homologous points in the left and right images are related by disparity (Fig. 2): $(x_R, y_R) = (x_L, y_L) + (u, v)$ (we will omit the subscripts from now on),

- the homologous points have similar intensities:

$$I_L(x, y) \approx I_R(x + u, y + v) \quad (1)$$

- the disparity vectors at neighboring spatial positions are similar (an accurate model within objects but somewhat imprecise at object boundaries),
- the camera geometry is close to parallel, thus inducing only small vertical disparities.

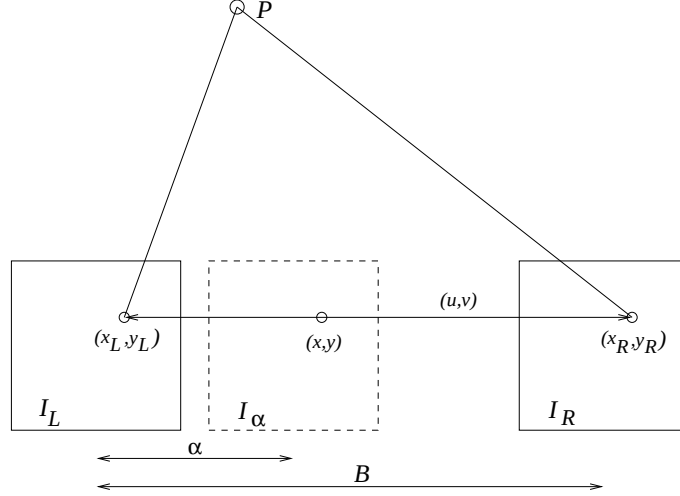


Figure 2. Homologous points in the left and right images (projections of a 3-D point P onto image planes). If virtual camera were present at position α , (x, y) would be another homologous point.

A slight modification of equation (1) is needed. Suppose that I_α is an image acquired by a virtual camera at baseline position α (Fig. 2). Let's assume that the camera separation B equals 1 for simplicity. Then, for $\alpha=0$ we have I_L and for $\alpha=1$ we have I_R . In order to reconstruct a pixel at spatial position (x, y) within the image I_α , the disparity vector (u, v) passing through (x, y) is needed. This vector needs to be pivoted, i.e., pass through, (x, y) and α , however its ends may move freely in I_L and I_R . In this way, if we know the true disparity vector (u, v) , we know the homologous points in images I_L and I_R to which (u, v) points, and thus we can recover the intensity/color of the pixel at (x, y) and α . Under these assumptions equation (1) can be re-written as follows:

$$I_L(x - \alpha u, y - \alpha v) \approx I_R(x + (1 - \alpha)u, y + (1 - \alpha)v). \quad (2)$$

Although the pivoting is not important in the case of continuous images, it is essential in practice when sampled images are used. Suppose that the disparity vectors were not pivoted at baseline position α , but elsewhere, e.g., at I_L or I_R . Then, the homologous points at α (intersections of disparity vectors with I_α - Fig. 2) would be located off the image sampling grid. Thus, a set of non-uniformly spaced points would be recovered and an additional step of non-uniform 2-D interpolation would be required, a highly non-trivial task.

Having defined the constant-intensity model mathematically, the goal is to find such disparities (u, v) that minimize the difference between both sides of equation (2). Since two parameters need to be computed for each pixel, the problem is underconstrained and additional assumptions, such as the similarity of neighboring disparities, mentioned above, need to be made.

Therefore, we minimize the error between both sides in (2) under regularization by disparity smoothness and a vertical disparity constraint:

$$\min_{(u,v)} \int \int [I_R(x + (1 - \alpha)u, y + (1 - \alpha)v) - I_L(x - \alpha u, y - \alpha v)]^2 + \lambda(u_x^2 + u_y^2 + v_x^2 + v_y^2) + \gamma v^2 dx dy \quad (3)$$

where u_x, u_y and v_x, v_y are partial derivatives of u and v , respectively. For simplicity of notation, the dependence of u and v on x, y is implicit.

In the minimization (3), the first term expresses the matching error for each disparity vector, the second term is a measure of consistence of a disparity vector with its neighbors (smoothness of the disparity field) and the third term is a measure of amplitude of the vertical disparity. The above formulation has been inspired by the work of March¹¹ and parallels Markov random field formulations used in motion estimation.¹² Compared to March's approach, our contributions are the extension to choose arbitrary pivot point for disparity vectors, the inclusion of vertical disparities and the introduction of vertical disparity constraint to force matching close to epipolar lines.

From the calculus of variations it is known that the above minimization is equivalent to solving the following Euler-Lagrange equations:

$$J(x, y, u, v)J_u(x, y, u, v) = \lambda \Delta u(x, y) \quad (4)$$

$$J(x, y, u, v)J_v(x, y, u, v) = \lambda \Delta v(x, y) + \gamma v(x, y) \quad (5)$$

where J is the displaced pixel difference:

$$J(x, y, u, v) = I_R(x + (1 - \alpha)u, y + (1 - \alpha)v) - I_L(x - \alpha u, y - \alpha v)$$

and J_u, J_v are its partial derivatives with respect to u and v , respectively:

$$\begin{aligned} J_u(x, y, u, v) &= (1 - \alpha)I_R^x(x + (1 - \alpha)u, y + (1 - \alpha)v) + \alpha I_L^x(x - \alpha u, y - \alpha v) \\ J_v(x, y, u, v) &= (1 - \alpha)I_R^y(x + (1 - \alpha)u, y + (1 - \alpha)v) + \alpha I_L^y(x - \alpha u, y - \alpha v). \end{aligned}$$

Above, $I_L^x, I_L^y, I_R^x, I_R^y$ are local partial derivatives of intensity at suitable spatial positions (compensated for disparity). They have been implemented based on the optimal bi-cubic interpolation proposed by Keys.¹³

We discretize the above problem in coordinates (i, j) and we approximate the Laplacian operator as follows:

$$\begin{aligned} \Delta u(i, j) &= 4(\bar{u}(i, j) - u(i, j)) \\ \Delta v(i, j) &= 4(\bar{v}(i, j) - v(i, j)) \end{aligned}$$

where $\bar{u}$ and $\bar{v}$ are disparity averages over the four nearest neighbors.

Now, that the Euler-Lagrange equations can be written in a discrete form, we solve them using Gauss-Seidel iterations as follows:

$$u^{n+1}(i, j) = \bar{u}^n(i, j) - \frac{1}{4\lambda} J(i, j, \bar{u}^n, \bar{v}^n) J_u(i, j, \bar{u}^n, \bar{v}^n), \quad (6)$$

$$v^{n+1}(i, j) = \frac{4\lambda}{4\lambda + \gamma} \bar{v}^n(i, j) - \frac{1}{4\lambda + \gamma} J(i, j, \bar{u}^n, \bar{v}^n) J_v(i, j, \bar{u}^n, \bar{v}^n). \quad (7)$$

Note that the right-hand sides of the equations have been made dependent on $\bar{u}^n$ and $\bar{v}^n$ rather than on u^n and v^n , as results from equations (4-5). This is a standard modification that improves stability of the system. Both iterative equations have the same form with the only difference being the weights; the second equation incorporates γ responsible for the constraint on vertical disparity amplitude.

As usual, λ and γ are weighting constants that need to be carefully selected for each image. This is, unfortunately, a drawback of the regularization-based methods; the final estimates depend on the λ and γ values.

4. SINGLE-VIEW RECONSTRUCTION

Having found either matching blocks or points in I_L and I_R *via* disparity estimation, the reconstruction of an intermediate image should be ideally as simple as copying either the corresponding left or right block/pixel to the intermediate image I_α . However, due to noise, illumination effects and various distortions (e.g., aliasing) the matching pixels, even if truly homologous, usually are not identical. This is even more true for blocks. Thus, to reconstruct I_α , we propose to use disparity-compensated linear interpolation with a simple two-coefficient interpolating kernel:

$$I_\alpha(i, j) = \mu_L I_L(i - \alpha u, j - \alpha v) + \mu_R I_R(i + (1 - \alpha)u, j + (1 - \alpha)v), \quad \mu_L + \mu_R = 1 \quad (8)$$

at all pixel positions (i, j) in the intermediate image I_α . Note that since the problem has been discretized, pixel positions (i, j) are now integer but disparity vectors (u, v) remain to be continuous (float); bi-cubic interpolation is used to recover sub-pixel intensities.¹³ The constraint on the sum of the coefficients assures a unit gain of the filter.

In general, good results have been obtained with the following selection of the coefficients:

$$\mu_L = 1 - \alpha, \quad \mu_R = \alpha$$

that assures high weighting of the close image and low weighting of the distant image. Note that such a filter is of low-pass type. This, however, does not work properly when parallax increase is required, i.e., for $\alpha < 0$ or $\alpha > 1$, since the coefficients μ_L and μ_R may take values smaller than 0 or greater than 1. For some values of α this will result in a filter with significant gain in high frequencies thus causing undesirable effects such as noise amplification.

Therefore, a normalized set of coefficients will be used to assure low-pass behavior of the temporal filtering process:

$$\mu_L = \frac{|1 - \alpha|}{|1 - \alpha| + |\alpha|}, \quad \mu_R = \frac{|\alpha|}{|1 - \alpha| + |\alpha|}.$$

For $0 < \alpha < 1$, the above definition results in the same weights as before, however for $\alpha < 0$ or $\alpha > 1$ the weights stay within the 0 to 1 range due to normalization.

The interpolation (8) works well if disparity is precisely estimated, otherwise the reconstructed images suffer from edge blurring due to averaging of misaligned edges. An alternative is disparity-compensated non-linear interpolation⁵ where pixels in I_α are optimally selected to come from *either* left *or* right image.

5. MULTIPLE-VIEW RECONSTRUCTION

Equation (8) describes how to compute an intermediate view at a single position α between I_L and I_R . However, for a continuous parallax adjustment or “continuous look-around” intermediate images at many positions α are needed.

5.1. Reconstruction from disparity sequence

One option is to estimate a disparity field at each α and reconstruct I_α using this disparity. Since the disparity estimation and the image reconstruction are performed at the same position, let’s call this approach the “optimal” IVR. For applications where many views are required this is a very costly solution (computationally).

5.2. Reconstruction from single disparity

An alternative to the computation of a disparity field at each baseline position is to compute a single disparity field at position α and then use it to reconstruct images at various positions ρ . Let $I_\rho(\alpha)$ denote an image reconstructed in this way - the subscript denotes where the image is reconstructed while the argument in parentheses denotes where the disparity is estimated. Clearly, $I_\alpha(\alpha)$ is the “optimally” reconstructed image where the disparity estimation and image reconstruction are performed at the same position α .

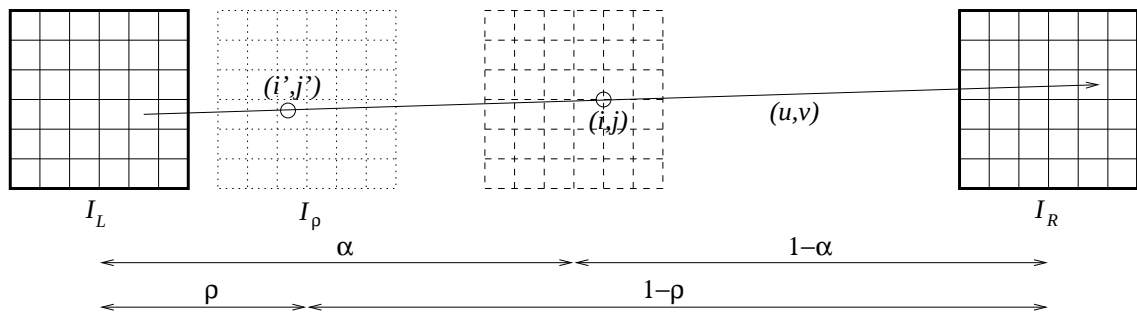


Figure 3. Disparity-compensated propagation of disparity vector from position α to position ρ . Disparity is computed at (i, j) and α , and its intersection point (i', j') (non-integer) with the image plane at ρ is found by disparity compensation. An almost epipolar geometry is assumed so that both α and ρ are measured along the line connecting optical centers of both cameras.

Clearly, to do this the disparity field at ρ needs to be inferred from the disparity field at α . One possibility is to perform disparity compensation, as shown in Fig. 3. Under the assumption of parallel camera geometry, a disparity

vector (u, v) applies to all pixels located at intersections of this vector with image planes at variable ρ 's. The difficulty is that at a position ρ different from α , the intersection points of all disparity vectors are not, in general, uniformly spaced. It is only at α , where the vectors are pivoted, that those intersections are uniformly spaced (in fact they coincide with the sampling grid). Therefore, a costly non-uniform interpolation is required. This is compounded by the fact that in some image areas few intersections may exist, thus significantly increasing a likelihood of interpolation errors.

Another alternative, that we adopt here, is a non-compensated propagation. Since in such applications as 3-D video-conferencing or 3-D TV stereoscopic images are encoded to exploit cross-view redundancy, disparity information for $\alpha = 0$ or $\alpha = 1$ is known at the receiver. Thus, it would be beneficial to perform a multiple- ρ IVR based on a single disparity field (for each I_L/I_R pair). Since we have assumed that spatially-neighboring disparities are similar, we also assume that they are similar α -wise (the same spatial position at neighboring α 's). Exploiting this assumption, we propagate (copy) the disparities computed at α to ρ (Fig. 4) and use them for interpolation (8). Note that copying the vector (u, v) to a new baseline position forces the use of ρ in (8) instead of α .

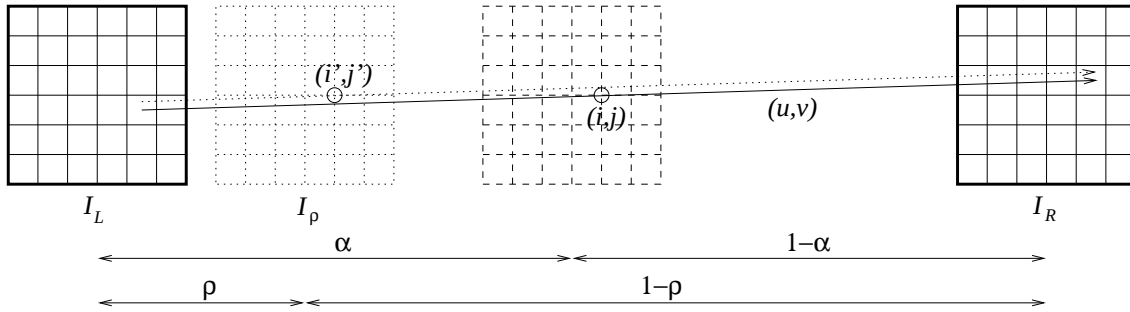


Figure 4. Non-compensated propagation of disparity vector from position α to position ρ . Disparity is computed at (i, j) and α , and then moved to position $(i', j') = (i, j)$ at baseline position ρ ; no-disparity compensation is performed. Solid-line vector denotes the estimated disparity, while the dotted-line vector denotes the propagated disparity. An almost epipolar geometry is assumed so that both α and ρ are measured along the line connecting optical centers of both cameras.

6. RECONSTRUCTION FOR INTERLACED DISPLAYS

The methodology described above applies directly to progressively-scanned stereo images subsequently evaluated on a progressive display. In the case of an interlaced display, an adaptation to even-field/odd-field image scan must be incorporated into the interpolation algorithm.

Let's define an indicator function ξ_κ that equals 0 if field κ is even and 1 if it is odd. Then, an intermediate field I_ρ is interpolated as follows (Fig. 5):

$$I_\rho(i, j) = \mu_L I_L(i - \rho u, j + 0.5(\xi_\rho - \xi_L) - \rho v) + \mu_R I_R(i + (1 - \rho)u, j + 0.5(\xi_\rho - \xi_R) + (1 - \rho)v)$$

where ξ_L , ξ_ρ and ξ_R are indicator functions for fields I_L , I_ρ and I_R , respectively, and μ_L , μ_R are functions of ρ . Note that, as shown in Fig. 5, the line count starts at the top of each image.

7. EXPERIMENTAL RESULTS

We applied both block- and pixel-based methods to several stereoscopic images (CCETT, NHK)* with two goals in mind: to compare the performance of the methods and, more importantly, to evaluate the usefulness of parallax adjustment as an enhancement for stereoscopic viewing systems.

On all images tested (moderate to high complexity) both algorithms performed very well. Not surprisingly, both methods failed on *Manege* that shows a very complex scene with some very narrow objects whose depth is difficult

*See the Acknowledgments section at the end of the paper.

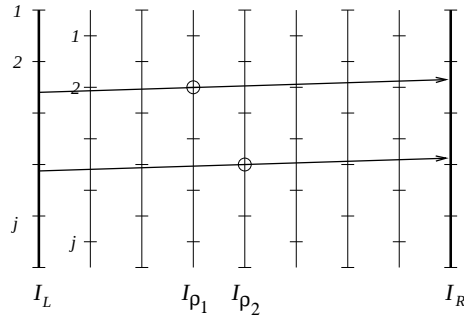


Figure 5. Intermediate view reconstruction for an interlaced display. Vertical intersection of each field is shown; lines are counted from the top. Each field corresponds to a position ρ_κ .

to estimate. Overall, the pixel-based approach has been judged to produce better results due to gradual distortions at object boundaries and in occlusion areas rather than block structure visibility present in the block-based results. The differences were quite evident when reconstructed images were played one after another to simulate camera pan, but much harder to discern once examined still. Fig. 6 shows a window from images reconstructed by both methods. Note the blocks on the boundary between the flowers and the building wall.

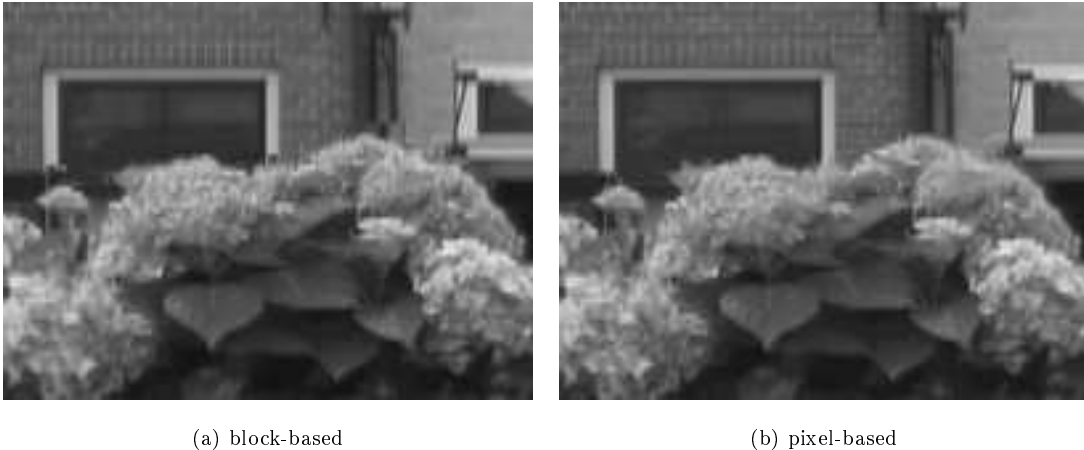


Figure 6. 200×150-pixel window from image *Flower* reconstructed at $\rho=0.5$ by: (a) block-based reconstruction, and (b) pixel-based reconstruction.

Next, we compared the performance of the “optimal” IVR (Section 5.1) with the single-disparity IVR (Section 5.2). In a subjective evaluation of individual images reconstructed by both methods viewers found very small differences but could not express preference as to which image was better. To demonstrate how small the differences are, Fig. 7 shows a magnified difference between images reconstructed using both methods.

Since it is difficult to examine all images created by various combinations of α and ρ , we computed a mean-square error between the images $I_\alpha(\alpha)$ and $I_\rho(\alpha)$ for 16 different values of α and ρ (between -0.25 and 1.25). The resulting surface (Fig. 8) shows the discrepancy between the optimal case and the one with disparity propagation. A few interesting observations can be made. First, note that no error is introduced on the diagonal line ($\alpha=\rho$) since in this case no propagation is performed.[†] Secondly, note that for a fixed α (where disparity is estimated) there are two minima: at $\rho=0$ and at $\rho=1$. This is logical since the estimated disparities have little impact on image reconstruction close to one of the true images. The maximum around $\rho=0.5$ is especially pronounced for α around 0 and 1 since

[†]In the plot, the error on the diagonal is non-zero since we added a vertical offset to reflect the fact that even the optimal case is not error free.

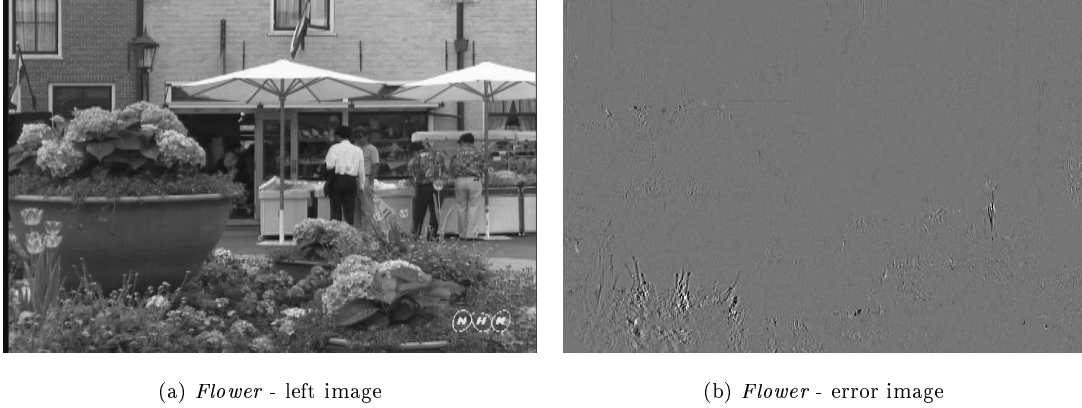


Figure 7. (a) The original left image from *Flower* and (b) the difference between luminances of two *Flower* images reconstructed at $\rho=0.1$: optimal image (disparity computed at $\alpha=0.1$) and sub-optimal image obtained by disparity propagation (disparity computed at $\alpha=0.5$). The difference is scaled by 5 and offset by 128 as follows: $5 \cdot (I_\alpha(\alpha) - I_\rho(\alpha)) + 128$.

this is the largest discrepancy between disparity position and reconstructed image position for the sub-optimal case. Note that for $\alpha=0.5$ the surface is “flatter” than for other α ’s suggesting a good position for estimating the disparity field. Thirdly, for a fixed ρ (reconstruction position) there is relatively little increase in the error until the disparity estimation position α moves outside the (0,1) range. This means that an image can be reliably reconstructed from any disparity computed within the (0,1) range.

These results, although computed on a small set of images, are very encouraging since a single disparity field may give base for the reconstruction of multiple images thus resulting in very large computational savings (disparity estimation is about two orders of magnitude more complex computationally than the reconstruction). Instead of computing, e.g., 20-30 disparity fields, 1 or perhaps 2 may be enough.

Since all the images used here come from interlaced cameras, in some of the tests we have reconstructed not progressive but interlaced sequences of intermediate images, as shown in Fig. 5. Played at 50-60 fields per second these sequences did not reveal any particular characteristics, but in slow motion, i.e., at around 10 fields per second, we noticed that some fields look sharper than others. In particular, for even stereo pairs (I_L, I_R) , it was the even fields that looked sharp and the odd fields looked blurred, especially vertically. This was caused by the fact that the vertical disparity is close to zero throughout the image (almost parallel cameras) and thus even fields require little vertical interpolation at I_L and I_R (small offset from the sampling grid). At the same time the odd fields require maximum vertical interpolation as disparity vector’s endpoints are positioned almost in the middle between two consecutive lines in I_L and I_R (Fig. 5). The problem cannot be easily solved in practice since parallel or almost parallel cameras are likely to be dominant in many applications, while high-order vertical interpolation is too costly and may not solve the problem completely. Clearly, interlaced displays should not be used for critical-quality stereo applications requiring disparity compensation if camera geometry is parallel.

To illustrate the impact of parallax adjustment on images at various ρ ’s, Figs. 9 and 10 show small windows extracted from *Flower* and *Piano* always at the same spatial position; the object displacements are too small to be shown on the whole image. Note that while in Fig. 9 the tulips move outwards with ρ approaching 0.5, in Fig. 10 the leaves move inwards with the same change in ρ . This is caused by the fact that the tulips in Fig. 9 induce crossed parallax on the screen, whereas the leaves in Fig. 10 cause an uncrossed parallax.

In informal subjective viewing, both parallax reduction and parallax increase were shown to a number of viewers. All viewers easily perceived the adjustment of depth and, except for some distortions at object boundaries, judged the quality to be very good. In terms of the overall usefulness of depth adjustment, the tests have shown a very favorable response. The overall comment was that the perceived depth varied smoothly from zero (one camera) to an exaggeration (cameras farther apart). The effect was judged very convincing and interesting. In fact, remarks have been made to the effect that this would make a “cool” special effect in a 3-D program. More tests are planned for the near future.

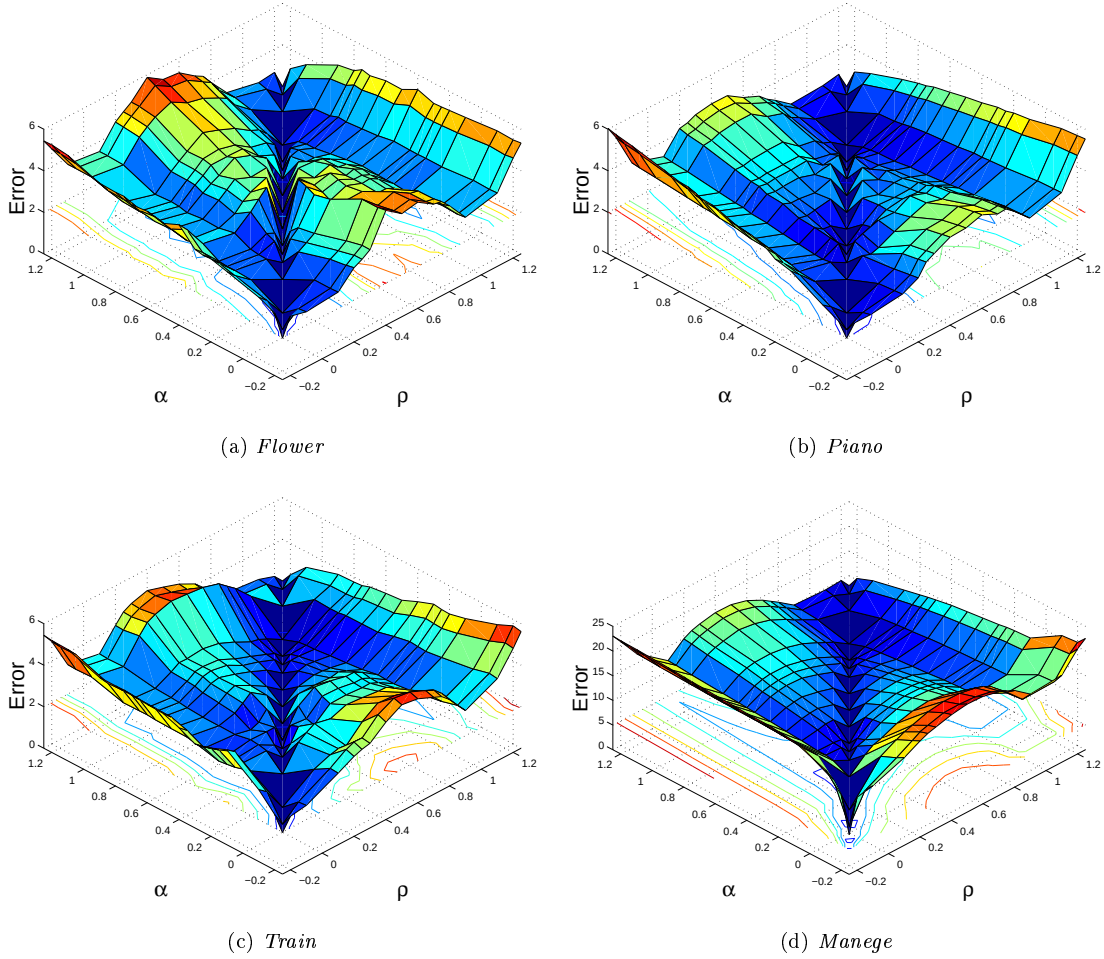


Figure 8. Mean-square error of the difference between views reconstructed using the optimal approach and the propagation. α denotes the position at which disparity to be propagated was computed, whereas ρ denotes the position at which the view was reconstructed. In each plot the surface was offset vertically to reflect the fact that even the optimal IVR induces reconstruction errors.

8. CONCLUSIONS

A necessary condition for the success of stereoscopic and 3-D applications in the consumer market is viewer comfort. All too often the first-time experience with stereo images is the last one due to eye strain, head discomfort (eyewear), etc. In this paper I have described a possible step in the direction of reducing viewer discomfort by adjusting the amount of screen parallax. This may be useful for users who are hypersensitive to depth cues or when stereo images exaggerate such cues. The presented parallax adjustment has been favorably evaluated by a number of viewers and found very believable. Hopefully, in the future all 3-D TV and computer screens will be equipped with a “depthness” or “3D-ness” knob in addition to brightness and contrast adjustments.

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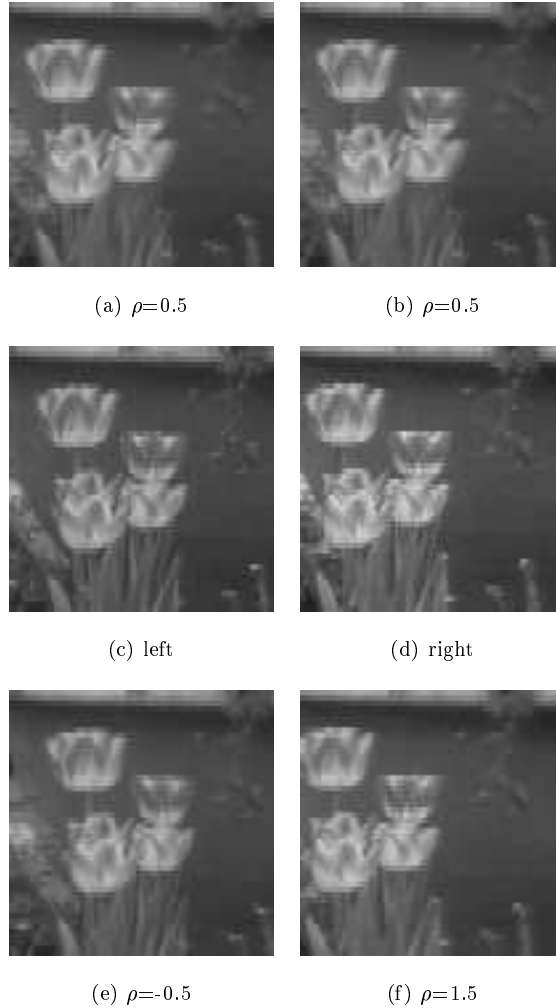


Figure 9. 100×100 -pixel windows extracted from *Flower*: (a-b) reconstructed images with parallax reduction, (c-d) original images, (e-f) reconstructed images with parallax increase. Note that in the top row the images are identical ($\rho=0.5$) and therefore no depth can be perceived.

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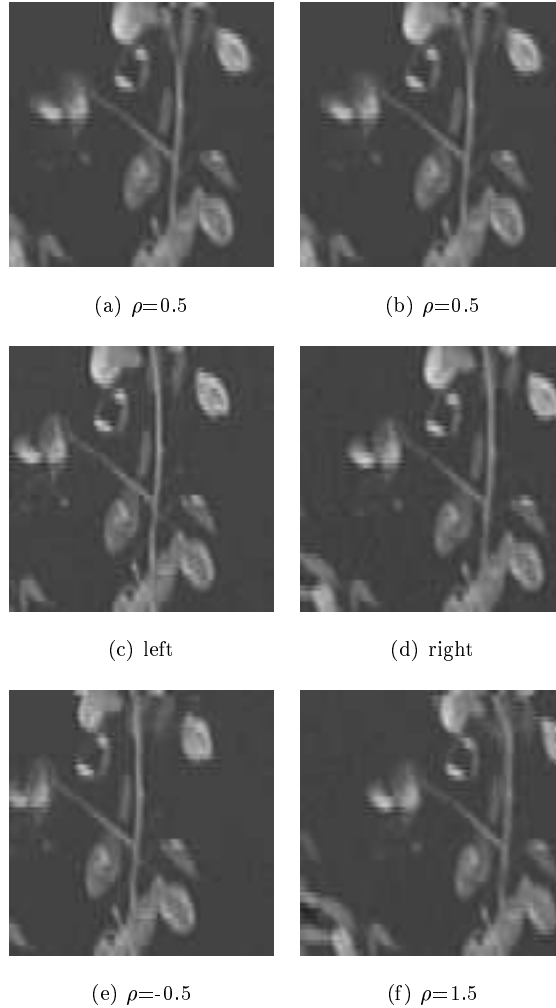


Figure 10. 100×100-pixel windows extracted from *Piano*: (a-b) reconstructed images with parallax reduction, (c-d) original images, (e-f) reconstructed images with parallax increase. Note that in the top row the images are identical ($\rho=0.5$) and therefore no depth can be perceived.

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